Mihai Zegheru

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Experience

Site Reliability Engineer Intern, Google, *Disaster Resilience* *July 2026 – October 2026*

- Led an end-to-end breakage prevention initiative for Google's foundational disaster recovery infrastructure, translating broad product requirements into observability and validation for critical files and packages required during fleet wide recovery.
- Designed and built a metrics pipeline and prefix tree validator across the majority of Google's repository to detect missing and incorrectly included content; uncovered major infrastructure bugs and multiple invalid sync rules affecting recovery completeness and content security rules. Added best-effort policy filtering for satisfying different constraint.
- Designed and built a security compliant ssh probe for SRE infrastructure, integrating it with critical user journey monitoring to extend existing SSH availability checks after an OpenSSH update disruption.

Software Engineer Intern, Google, *Core* *July 2025 – September 2025*

- Built an AI-powered automation used by 300+ legal experts, achieving 4x higher accuracy than the previous approach and outperforming human evaluators on an open-ended scoring task.
- Automated benchmarking and prompt generation, accelerating iteration and enabling fast rollout into shadow testing.
- Extended an AI governance platform by improving its design and building supporting API infrastructure, collaborating across 3 teams.

STEP Intern, Google, *BigQuery* *July 2024 – September 2024*

- Developed a metrics system for SQL Algebra Tree execution flow using RAII pattern, enabling data-driven development for the whole team.
- Owned design and implementation of core system architecture, prototyped and led development.

Education

Politehnica University of Bucharest *2023 – 2027*

Bachelor of Science, Computer Science and Engineering, GPA: 9.1/10 Graduation: July 2027

- Relevant Coursework: Operating Systems, Algorithms & Data Structures, Computer Networks, Computer Architecture, Probability & Statistics, Optimization Theory, Numerical Methods

Awards

National Olympiad of Computer Science, *Bronze Medal and 3rd Regional Prize* *2020 – 2023*

ICPC, *National phase* *2024, 2025*

Projects

athreads: Preemptive MCU Scheduler, *GitHub*

- Bare-metal preemptive threading library for the ATmega2560 in C/AVR assembly, implementing Timer1 driven round-robin scheduling, register and stack preserving context switches, sleeping threads, and per-thread time quanta.
- Added per-thread CPU accounting and USART telemetry, streaming live scheduler utilization to a Python visualizer.

Sable: Data-Oriented ECS Game Engine, *GitHub*

- C++23 game engine with a data-oriented ECS, 16 KB archetype chunks, compact entity IDs, bitset signature queries, and prioritized systems for cache friendly iteration and predictable update phases.
- Implemented an OpenGL 4.6 renderer with Assimp model import, explicit GPU resource uploads, bucketed shader draw submission, UI, and render to texture camera passes.

Market Simulator, *GitHub*

- Built a C++20 order book engine with price-time priority, partial fills, cancel/replace semantics, server assigned order IDs, and per-order trade attribution.
- Implemented TCP order entry using epoll, TCP_NODELAY, and a length prefixed binary protocol, plus a UDP multicast feed for one-to-many order book snapshots.

Router Dataplane, *GitHub*

- Built a C router data plane implementing IPv4 forwarding, dynamic ARP resolution, and ICMP echo and error handling over Ethernet.
- Implemented a binary trie for longest-prefix matching in at most 32 bitwise steps, with packet queuing until next-hop MAC addresses are resolved.

Custom Memory Allocator, *GitHub*

- Implemented malloc/calloc/realloc/free equivalents in C using brk/mmap, with 8-byte alignment, block splitting, coalescing, and threshold-based allocation.

Skills

Programming Languages: C++, C, Go, Python, Java, Assembly, Bash, SQL

Technologies: Valgrind, GDB, Git, Linux, OpenGL, AVR/ATmega microcontrollers